Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/07/2020
PI Name:	Toscano, William Ph.D.		
Project Title:	Evaluation of an On-Astronaut Wireless Sensor System to Support Crew Health Monitoring for Exploration Class Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	william.b.toscano@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-2324
Organization Name:	NASA Ames Research Center		
PI Address 1:	Human Factors Research and Technology Division		
PI Address 2:	Mail Stop 262-2		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	10/01/2018	**End Date:**	10/01/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
Contact Email:	jay.lemery@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Final report: Added Co-Investigator: Roxanne Buxton, Wyle Labs, NASA Johnson Space Center.		
COI Name (Institution):	Cowings, Patricia Ph.D. (NASA Ames Research Center) Shaw, Tianna M.S.B.M.E. (NASA Ames Research Center) Hanson, Andrea Ph.D. (NASA Johnson Space Center) Krihak, Michael Ph.D. (Universities Space Research Association/NASA Ames Research Center) Martin, Annie Ph.D. (Canadian Space Agency) Buxton, Roxanne E M.S. (Wyle Labs, NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA and our international partners have a mutual interest in advancing biosensor monitoring capabilities and medical data management to support crew health and performance during future exploration missions. Studying this technology on Earth may identify ways to preserve and advance crew fitness levels during preflight training, on-orbit operations, and postflight rehabilitation. In addition, crew monitoring will provide critical data in the diagnosis and treatment of medical conditions, as well as aid in the evaluation of crew fitness, health, and readiness for mission operations. For an exploration mission NASA needs a health monitoring system composed of hardware that is compact, fully interoperable with an integrated data management system, and requires minimal consumables. Such a system will be achieved through the integration of small, easy to use biomedical sensors that will have the ability to measure, store, and transmit physiologic parameters during operational and ambulatory scenarios. Among the parameters of interest are key physiological signals that indicate crewmember workload and relevant vital signs including ECG (electrocardiogram), heart rate, and respiration rate. Since 2015 NASA Ames Research Center (ARC) and the Canadian Space Agency (CSA) have been collaborating on studies to evaluate a shirt-based biosensor system, referred to as Astroskin, for continuous vital sign monitoring. Lessons learned from previous test activities led CSA to engage in technology development to explore the capabilities of a wireless version of the Astroskin utilizing small, unobtrusive, stand-alone patch sensors. CSA's prototype On-Astronaut Wireless Sensor System (OAWSS) will measure and transmit real-time multiple physiological parameters. In order to successfully advance the Technology Readiness Level (TRL) of biosensor monitoring devices, it is essential to determine whether the equipment works appropriately by providing reliable and valid data under various activity levels. Two independent studies were carried out to evaluate OAWSS performance and to help validate its operational feasibility for astronaut health monitoring during future exploration missions. Furthermore, since engaging in physical activities similar to normal crew activities is likely to affect the signal quality of recorded physiological data largely due to motion artifact, quantifying the percent of time that unusable signals are generated in various levels of activity will enable manufacturers to improve and enhance current technologies in order to better meet stakeholder needs. Specific Aims: 1. To evaluate the reliability and validity of OAWSS measures during controlled laboratory tests that include a range of physical activities, an operational flight-like aerobic exercise test, and a simulated EVA task. 2. To determine whether any of the OAWSS sensors provide acceptable data as defined by the percentage of time a usable signal is detected during lab tests. 3. To assess wireless data connectivity of OAWSS sensors during 48-hour of monitoring that include normal daily activities while at work and home. 4. To evaluate participant comfort, interference with daily activities, and functionality of the OAWSS from participant surveys. Methods: The proposed research was conducted in two phases. In a Phase 1 study at Ames Research Center OAWSS measures were collected on 9 participants (5 men and 4 women) during laboratory tests that included low-intensity physical exercise on a treadmill, a cycle ergometer, and a seated rower. In addition, participants wore the OAWSS sensors for a maximum of 48 hours during normal activities outside the lab to assess long-term wear and comfort. During lab tests measures were acquired with standard laboratory equipment in parallel with the OAWSS measures. A Phase 2 study was conducted at Johnson Space Center with 8 participants (4 men and 4 women) to further demonstrate the capabilities of the OAWSS during simulated exploration activities that included a flight-like exercise protocol on a cycle ergometer, and a simulated extravehicular activity (EVA) task (hill climb in a weighted suit) on a Treadmetrix device. These tests were conducted with standard bio-monitoring devices for comparison to the OAWSS measures. All participants completed a survey questionnaire on usability, interference and comfort of the OAWSS system. Significance: The above test activities will help to validate the OAWSS operational feasibility for long-term wear and crew health monitoring.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Unobtrusive, wireless devices for monitoring vital signs have evolved over the past decade. The OAWSS is a prototype wireless skin patch sensor system developed by the Canadian Space Agency for monitoring crew health during future exploration missions. The OAWSS has potential value for out-patient health monitoring and monitoring patients in remote locations who have limited access to health care facilities. In addition, the OAWSS may be beneficial in the sports industry for training professional athletes.
Task Progress:	The On-Astronaut Wireless Sensor System (OAWSS), developed by the Canadian Space Agency (CSA), is a prototype, stand-alone patch sensor system designed to measure relevant vital signs of astronauts during future exploration class missions. This study evaluated the reliability and validity of the OAWSS for measuring heart rate (HR), respiration rate (RR), skin temperature (ST), and galvanic skin response (GSR) against criterion devices (e.g., EQ02 LifeMonitor, Equivital Inc., New York, NY and E4 wristband, Empatica, Boston, MA, respectively). In study 1 five male and four female volunteers (mean ± SD, 39.1 ± 5.1 yr) completed two days of testing in which heart rate (HR), respiration rate (RR), skin temperature (ST), and galvanic skin response (GSR) were collected from the OAWSS and criterion devices at the same time for approximately 90 minutes while engaged in low, medium, and high intensity activities which were repeated on two consecutive days. In study 2 four male and four female volunteers (mean ± SD, 35.1 ± 6.5 yr) completed one day of testing in which the above parameters were collected for approximately 3 hours while engaged in a flight-like aerobic exercise protocol and a simulated extravehicular activity (EVA) task. Reliability was determined by calculating the change in measured parameters (i.e., error scores) between test activities on day 1 and day 2 (study 1 only). Validity was determined by calculating the change in measured parameters between OAWSS and criterion devices during test activities of both studies 1 and 2. Data were analyzed using a 2-way repeated measures ANOVA and Bland-Altman plots showed the dispersion of the individual reliability and validity error scores. Other statistics were calculated i.e., prediction intervals and correlation coefficients and used as criterion measures for reliability and validity. Results demonstrated that only OAWSS ST was reliable during activities in study 1, and only OAWSS HR hill climb met the validity criteria in study 2. In conclusion, the OAWSS should not be evaluated further in its current configuration based on the poor reliability and validity of measurements from both studies. In addition, the low acceptability results strongly suggest the OAWSS will need a significant form factor improvement in order to be acceptable to flight crew as a health monitoring device. Future research should also examine other physiological variables such as body position, movement, SPO2, and blood pressure, to more accurately determine health status.
Bibliography Type:	Description: (Last Updated: 07/07/2020)

Abstracts for Journals and Proceedings	Toscano W, Cowings P, Sullivan P, Martin A, Krihak M, Winther S, Hanson A, Shaw T. "Wearable biosensor monitor to support autonomous crew health and readiness to perform." Presented at the 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. Abstracts. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017